

Lushness of vector-valued L_1 : a literature-implied answer

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Status. `literature_implied_answer` (full).

Source question

Choi, Kim, Lee, and Martín ask on page 7, immediately after Corollary 2.13 of [1]:

Whether $L_1(\mu, X)$ is lush for every lush space X is not known to the best of our knowledge.

The surrounding results allow an arbitrary measure μ .

Supporting theorem

Kadets, Martín, Merí, and Pérez prove in [2, Theorem 8.10(b) and Corollary 8.12, p. 92] that, for every sigma-finite measure ν ,

$$X \text{ is lush} \iff L_1(\nu, X) \text{ is lush.}$$

Their Remark 8.11 says that the lushness assertion is new even for the identity operator.

Identification in the arbitrary-measure case

Proposition. *Let (Ω, Σ, μ) be an arbitrary measure space. If X is lush, then $L_1(\mu, X)$ is lush.*

Proof. Fix $f, g \in S_{L_1(\mu, X)}$. Up to null sets, put

$$A = \{\omega : \|f(\omega)\| + \|g(\omega)\| > 0\}.$$

This set is sigma-finite: writing

$$A_n = \{\omega : \|f(\omega)\| + \|g(\omega)\| \geq 1/n\},$$

we have

$$\mu(A_n) \leq n(\|f\|_1 + \|g\|_1) = 2n, \quad A = \bigcup_{n \geq 1} A_n.$$

Consequently $Z = L_1(\mu|_A, X)$, identified with the closed subspace of functions supported in A , is lush by Corollary 8.12 of [2].

Given $\varepsilon > 0$, apply lushness in Z to f, g . It gives $z^* \in S_{Z^*}$ such that f belongs to the relevant ε -slice of B_Z and g lies within ε of its absolutely convex hull. Extend z^* to $\tilde{z}^* \in S_{L_1(\mu, X)^*}$ by Hahn–Banach. The slice of B_Z is contained in the corresponding slice of $B_{L_1(\mu, X)}$, so the same witness works for f, g in the ambient space. Since f, g and ε were arbitrary, $L_1(\mu, X)$ is lush. $\square$

Provenance and scope

The sigma-finite theorem is explicit in the supporting source. The passage to an arbitrary measure via the joint sigma-finite support of two Bochner-integrable functions is the direct identification recorded here. Thus the entire forward implication asked in [1] is settled, but the provenance is a known theorem plus a short reduction, not a new result of this run.

References

- [1] Y. S. Choi, S. K. Kim, H. J. Lee, and M. Martín, *On Banach spaces with the approximate hyperplane series property*, Banach J. Math. Anal. 9 (2015), no. 4, 243–258; [arXiv:1407.7848](#).
- [2] V. Kadets, M. Martín, J. Merí, and A. Pérez, *Spear operators between Banach spaces*, Lecture Notes in Mathematics 2205, Springer, 2018; [arXiv:1701.02977](#).